

AI and ML Models in Sentiment E-commerce Platform

Generated on May 04, 2025

Introduction

This document provides detailed information about the AI and ML models used in the Time-Based Sentiment Analysis feature of our Sentiment E-commerce Platform. It includes the models used for analyzing sentiment trends and the parameter tuning methods employed to optimize their performance for e-commerce review data.

1. Overview of AI and ML Models

The following AI and ML models are currently implemented in the Time-Based Sentiment Analysis feature:

VADER (Valence Aware Dictionary and sEntiment Reasoner)

A lexicon and rule-based sentiment analysis tool specifically attuned to sentiments expressed in social media. NLTK package is used to implement VADER.

Time Series Forecasting

Statistical time series analysis models to predict future sentiment trends based on historical data patterns.

Anomaly Detection Algorithm

Identifies significant sentiment shifts that deviate from normal patterns. Helps detect sudden changes in customer sentiment that might require attention.

Seasonal Decomposition

Algorithms to separate sentiment data into trend, seasonal, and residual components, allowing identification of recurring patterns across different time periods.

K-means Clustering

Used in the Product Category Breakdown to group similar products by sentiment patterns.

Correlation Analysis

Statistical correlation methods to find relationships between product pricing and sentiment scores.

2. Parameter Tuning Methods

Each model has been carefully tuned to optimize performance specifically for e-commerce sentiment analysis. Below are the tuning methods applied to each model:

VADER Sentiment Analysis Tuning

Threshold Optimization: Standard VADER thresholds (typically 0.05 for neutral) have been adjusted to better categorize Amazon product reviews. Our tuned thresholds use 0.1 for positive sentiment and -0.05 for negative sentiment based on analysis of actual Amazon review data.

Domain-Specific Lexicon Enhancement: VADER's base lexicon has been augmented with e-commerce specific terms (like "shipping", "delivery", "refund") and their sentiment values based on empirical analysis of Amazon reviews.

Time Series Forecasting Tuning

Hyperparameter Optimization: Using grid search with cross-validation to find optimal window sizes, seasonality periods, and trend components.

Model Selection: Tested ARIMA, Exponential Smoothing, and Prophet models on historical review data to identify which provides the most accurate predictions for different product categories.

Anomaly Detection Parameter Tuning

Sensitivity Thresholds: Calibrated to identify true sentiment shifts while minimizing false positives.

Statistical Significance Testing: Implemented t-tests with p-value thresholds of 0.05 to ensure detected shifts are statistically significant.

Moving Window Size: Optimized to 14-day windows for short-term shifts and 30-day windows for longer-term trend detection.

Seasonal Decomposition Tuning

Seasonality Period Detection: Used auto-correlation analysis to identify the natural seasonality periods in the data rather than assuming fixed periods.

Decomposition Method Selection: Compared STL, X-12-ARIMA, and SEATS methods on Amazon review data to determine the most effective approach for e-commerce seasonality patterns.

K-means Clustering Tuning

Optimal Cluster Count: Used the elbow method and silhouette analysis to determine the optimal number of clusters for product categories.

Feature Selection: Selected the most informative features for clustering through feature importance analysis.

Correlation Analysis Tuning

Correlation Method Selection: Compared Pearson, Spearman, and Kendall correlation methods to find the most appropriate for non-linear price-sentiment relationships.

Confidence Intervals: Implemented bootstrap resampling to establish 95% confidence intervals around correlation estimates.

Conclusion

The Time-Based Sentiment Analysis feature combines multiple AI and ML models to provide comprehensive insights into customer sentiment trends. Each model has been carefully tuned to optimize performance specifically for e-commerce review data. The system aggregates review data from MongoDB, applies these various models to extract insights, and then visualizes the results through interactive charts. The ML pipeline is designed to continuously update as new review data comes in, providing real-time insights into customer sentiment trends.